

RUBBLEJACK · PRECIS

RubbleJack design precis

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DOCUMENT CONTROL

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Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
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Source	docs/02-concept.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.3	2026-10-03	Amish Chadha	TRL 3; numbers from RBJ-CAL-001 and the constructable design (RBJ-DDR-002); media regenerated from the model
0.2	2026-10-03	Amish Chadha	Populated to TRL 2 with how it works, components, key design choices (RBJ-DDR-001), first-order numbers, safety and concept media
0.1	2026-09-30	Amish Chadha	Initial scaffold

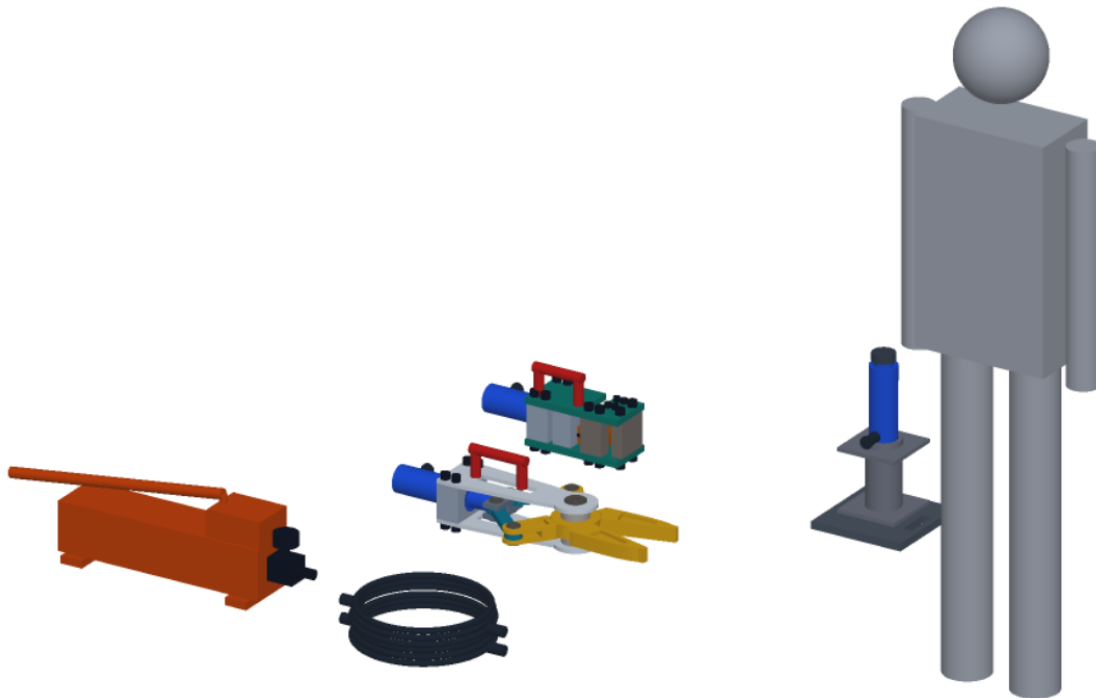
RubbleJack design precis

Turns commodity construction hydraulics into an open spreading, lifting and cutting kit for crews without powered rescue tools.

RubbleJack

github.com/BoujeeEnjinia1701/rubblejack

CONCEPT, NOT FOR FABRICATION



Seen from the front right and above, 24 deg elevation. Grey figure: 1.75 m person for scale

Figure 1. The kit as packed for a drill: hand pump with gauge and hoses, spreader, cutter and lifting ram set. CONCEPT, NOT FOR FABRICATION.

Safety: RubbleJack is safety-critical. It lifts and spreads loads of up to 142 kN (14.5 t) with oil at 700 bar (10,000 psi), in collapsed structures that can move. It is published as an open engineering reference, never as certified rescue equipment, and it is not certified to NFPA 1936, NFPA 1960 or EN 13204. See the Safety section below before reading further.

■ How it works

One person works a two-speed hand pump; a 2 m hose with quick couplers carries oil at up to 700 bar to whichever tool is connected. Each of the three tools has its own bought 15 t, 101 mm stroke single-acting cylinder, so changing tools means only uncoupling one hose and coupling another (RBJ-DDR-001, D2):

- **Spreader.** The cylinder pushes a crosshead forward; two links push the tails of two jaw arms that turn on one pivot pin, so the jaw tips open from 32 mm to 262 mm with 49 to 64 kN between them. The arms are held between two aluminium frame plates that bolt to a collar adapter screwed onto the cylinder's collar thread.
- **Cutter.** The cylinder pushes a moving blade past a fixed blade, like a guillotine. The bar lies in a hook cut in two steel plates and in a U notch in each blade; the 142 kN of the cylinder shears 16 mm reinforcing bar, which needs about 105 kN. A stop bar ends the stroke after the cut.
- **Lifting ram.** The third cylinder stands in a ring on a 250 mm steel base plate, with a tilting saddle on its plunger. A 150 mm riser block can go under it, never on top of the plunger, for 527 mm of reach. The crew cribs the load with hardwood as it rises.

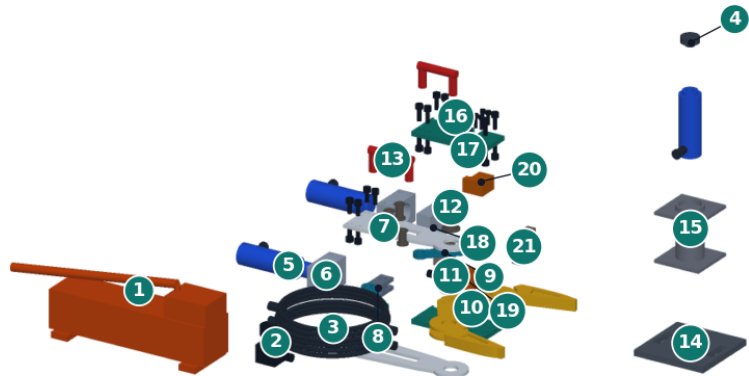
The gauge on the pump outlet shows the working pressure; the pump's internal relief valve stops at 700 bar. The cylinders retract by their own springs when the pump's release valve is opened, so the spreader and cutter have no powered closing force.

RubbleJack: exploded view

github.com/BoujeeEnjinia1701/rubblejack

CONCEPT, NOT FOR FABRICATION

- 1 Two-speed hand pump
- 2 Gauge adapter and pressure gauge
- 3 Hoses with quick couplers
- 4 Tilting saddle
- 5 Single-acting cylinder 15 t
- 6 Collar adapter
- 7 Spreader frame plate
- 8 Crosshead
- 9 Link
- 10 Jaw arm
- 11 Pivot spacers
- 12 Pins with washers and circlips
- 13 Carry handle
- 14 Ram base plate
- 15 Riser block
- 16 Cap screws, M16 and M12
- 17 Cutter plate
- 18 Blade carrier
- 19 Moving blade
- 20 Fixed blade
- 21 Nose block and stop bar



Seen from the front right and above, 24 deg elevation; numbers match bom/bom.csv

Figure 2. Exploded view; numbers match the BOM lines in Table 1.

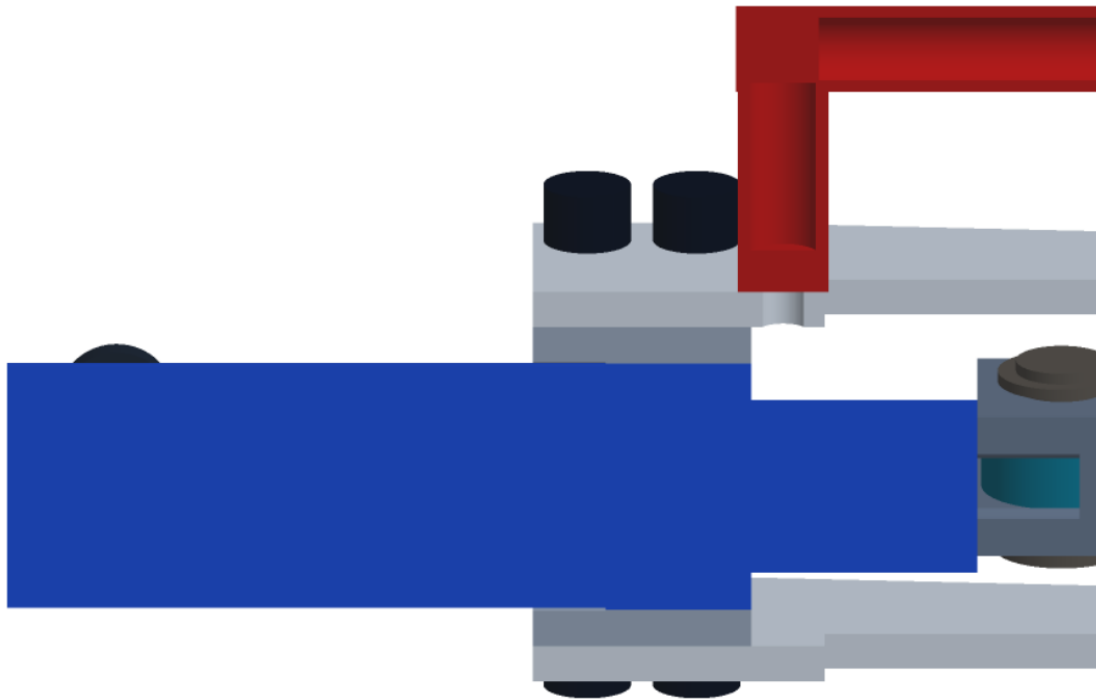
■ Components

Table 1. Main components (numbers match bom/bom.csv and Figure 2).

#	Component	Role	Made or bought
1	Two-speed hand pump, 700 bar	Powers every tool; fast stage at low load, slow stage at high pressure	Bought
2	Gauge adapter and gauge	Shows working pressure on the pump outlet	Bought
3	Hoses, 2 m, with quick couplers	Connect the pump to any tool without tools	Bought
4	Tilting saddle	Spreads the ram's push on an uneven load	Bought
5	Single-acting cylinder, 15 t (three)	One per tool	Bought
6	Collar adapter (two)	Joins a tool's frame plates to its cylinder's collar thread	Made, 7075-T6
7	Spreader frame plates (two)	Carry the pivot pin and the reaction back to the adapter	Made, 7075-T6 10 mm
8	Crosshead	Turns the plunger's push into two link pushes	Made, 7075-T6
9	Links (two)	Push the jaw arm tails	Made, S690QL
10	Jaw arms (two)	Spread the load at the tips	Made, 4340 quenched and tempered
11	Pivot spacers (two)	Fill the gap between arms and plates	Made, 7075-T6
12	Pins	Main pivot pin and four link pins	Made, 4340 quenched and tempered
13	Carry handles (two)	One on the spreader, one on the cutter	Made, steel tube
14	Ram base plate	Spreads the ram's load on cribbing	Made, S690QL 20 mm
15	Riser block	Extends the ram's reach by 170 mm	Made, S355 tube and plate
16, 24	Cap screws, M16 and M12	Hold every bolted joint	Bought
17	Cutter plates (two)	Hold the blades, nose and stop; the hook takes the bar	Made, S690QL 16 mm
18	Blade carrier	Carries the moving blade on the plunger	Made, 7075-T6
19, 20	Moving and fixed blades	Shear the bar	Made, S7 tool steel, hardened
21	Nose block and stop bar	Take the fixed blade's thrust; end the cutter stroke	Made, 7075-T6 and S355
22	M24 studs	Hold the crosshead and carrier on their plungers	Bought
23	Carry bags (four)	One bag a load	Bought

RubbleJack: cutawaygithub.com/BoujeeEnjinia1701/rubblejack

CONCEPT, NOT FOR FABRICATION



Spreader only, front half removed through the pivot; seen from the front and above, 18 deg elevation

Figure 3. Cutaway of the spreader, front half removed: frame plates, spacers, interleaved jaw arm knuckles and the main pin.

■ Key design choices

All are decided under Amish's 2026-10-03 pre-approval and recorded in RBJ-DDR-001 and RBJ-DDR-002; the register is RBJ-DEC-001.

- **Commodity 700 bar hydraulics** (D1): the most widely stocked hand hydraulic class; at least two makers offer every bought part.
- **One cylinder per tool** (D2): a tool change is a coupler swap of about 20 s, at the cost of two extra cylinders.
- **Linkage spreader and guillotine cutter** (D3, D4): pinned plate parts a local workshop can make; the cutter is rated for 16 mm bar only.
- **Located ram stack** (D5): base ring, riser spigot and saddle, so nothing stacked can slide; never a riser on the plunger.
- **Strength rule** (D7): every made load-bearing part at least 1.5 on yield at 700 bar; the lowest is the jaw arm tail at the link slot, 1.60.
- **Four loads under 25 kg** (D10): high-strength steel where sections are small and 7075-T6 aluminium where parts are bulky.

■ First-order numbers

All at the pump's 700 bar relief pressure, from RBJ-CAL-001; first-principles paper estimates, nothing measured.

Table 2. Key figures.

Quantity	Value	Assumption
Force per cylinder	142 kN (14.5 t)	20.3 cm ² effective area at 700 bar
Spreading force at the tips	49 kN closed to 64 kN fully open	Pin friction ignored; linkage geometry from the model
Tip opening	32 mm closed, 262 mm open	101 mm stroke
Force to shear 16 mm bar	About 105 kN; ratio 1.36 to the cylinder	Shear strength 0.8 of a 650 MPa tensile strength (upper bound for B500B and A615 grade 60)
Ram reach	256 mm closed, 357 mm open, 527 mm on the riser	Saddle included
Pump strokes, full stroke under load	About 228 slow-stage strokes, 9 min	0.9 cm ³ a stroke, 25 strokes a minute (estimate)
Pump strokes, one cut	About 41, under 2 min	37 cm ³ to close the cut (estimate)
Loads, packed	Pump set 15.9 kg, spreader 24.8 kg, cutter 22.0 kg, ram set 23.7 kg	0.8 kg bag each
Lowest strength margin	1.60 (jaw arm tail)	Minimum yield values; conservative hand calculations
Cost	Value-engineering target: USD 3,500. Estimated cost of the constructable design: USD 5,895 (USD 2,395 over the target)	Indicative prices in bom/bom.csv

RubbleJack: energy for one full spreader stroke at 700 bar, kJ (RBJ-CAL-001 estimates)

github.com/BoujeeEnjinia1701/rubblejack

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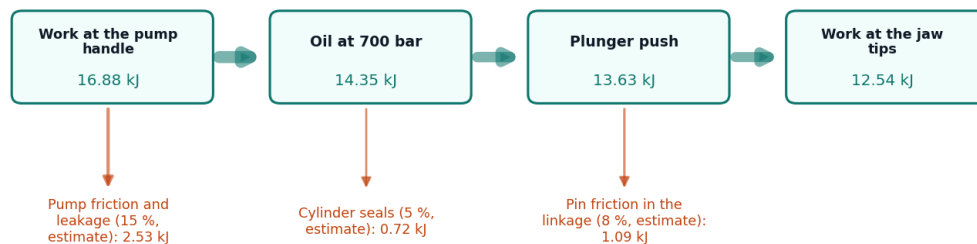


Figure 4. Energy for one full spreader stroke at 700 bar, from the pump handle to the jaw tips. All efficiencies are estimates.

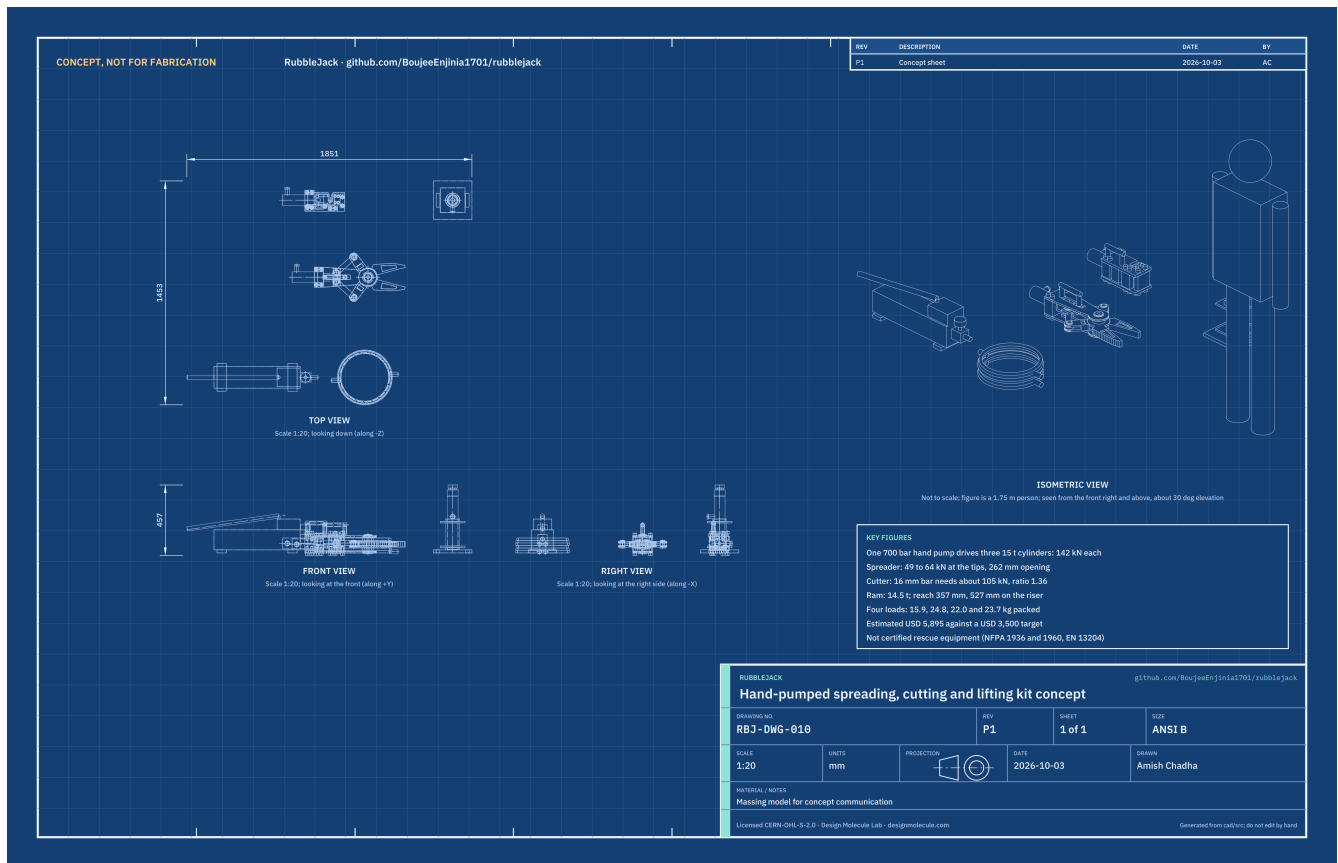


Figure 5. Concept blueprint RBJ-DWG-010 with orthographic views and key figures.

■ Patent design-arounds

From the preliminary patent, trademark and prior-art screen (not legal advice), kept as scaffolded:

- Use a commodity pump only, not a compact integrated pump unit (US11273547, Weber). RubbleJack uses a separate catalogue hand pump and hoses.
- Avoid features of the Holmatro portfolio, including the CC 23 crusher and its hand pumps. RubbleJack has no crusher and no integrated pump.
- Publish proof-load data and a clear not-certified notice naming NFPA 1936, NFPA 1960 and EN 13204 (RBJ-DDR-001, D8).
- Interchangeable tools on one hydraulic supply draw on lapsed prior art (US7937838B2); RubbleJack goes further and gives each tool its own cylinder.

■ Shared blocks

- Commodity hydraulics block: pump, hoses, couplers and gauge (adopted, RBJ-DDR-001, D13).
- CalRig: proof testing of the jaws, blades, ram stack and hoses at TRL 4.
- VoidScope: sibling project for searching the voids RubbleJack opens.

■ Safety

Safety: RubbleJack involves lifting, spreading and cutting with high-pressure hydraulics in unstable structures. It is not certified rescue equipment and must not be used on a real rescue on the strength of this document.

- **High-pressure oil.** Oil at 700 bar can pierce skin and cause injuries that need surgery. Never feel for leaks by hand; use card. Replace a damaged, kinked or abraded hose at once. Keep the gauge fitted; stop below 700 bar.
- **Lifting.** Never work under a load held only by a ram. Crib the load with hardwood as it rises and keep the cribbing within 25 mm of the load. Never lift on bare rubble; the base plate goes on hardwood sleepers. Never put the riser on top of the plunger.
- **Spreading and cutting.** Keep hands and bodies out of the jaws, the cutter hook and the line of any bar end. Cut bar ends and broken masonry can fly: eye and face protection for everyone near the tool. Cut only reinforcing bar of 16 mm or less, never hardened bar, chain, cable or springs.
- **Stored energy.** A spreading load or a bent bar holds energy; release the pump's valve slowly and stand clear of the line a load could move.
- **Structures.** Collapsed structures move in aftershocks. Use trained people, follow local incident command and shore before entering.
- **Weakest part.** Every part is rated for 700 bar; the pump's relief valve is the limit. Never fit a part rated below 700 bar. Each made tool is proof-loaded on CalRig and inspected for cracks before use (RBJ-DDR-001, D9).

■ Open questions

None. The questions from the scaffold were settled at the TRL 2 review (RBJ-DDR-001) and in the calculations (RBJ-CAL-001); facts that need real parts are listed in the design decisions register (RBJ-DEC-001) under "To confirm when parts are bought".